

1. 部分分式

將複雜的分式函數 $f(x) = \frac{P(x)}{Q(x)}$ 使用長除法與因式分解化簡成：

$$f(x) = S(x) + \frac{R(x)}{Q(x)},$$

其中 $S(x)$ 為多項式 (若 $\deg P \geq \deg Q$ 才有 $S(x)$ 這一項)，而 $\frac{R(x)}{Q(x)}$ 為真分式 ($\deg R < \deg Q$)。對於真分式，依據 $Q(x)$ 進一步拆為一次因式與不可分解的二次因式：

$$\frac{A_1}{x - a_1} + \cdots + \frac{A_m}{(x - a_m)^m} + \frac{B_1x + C_1}{x^2 + b_1x + c_1} + \cdots + \frac{B_nx + C_n}{(x^2 + b_nx + c_n)^n}$$

拆分後，每一項的積分通常簡單許多，從而簡化計算。

Question 使用部分分式拆解計算以下積分。

$$\begin{aligned} \triangleright \int \frac{2x^4 + 3x^3 + 7x^2 - 5x - 3}{x^3 - x} dx \\ = \int 2x + 3 + \frac{9x^2 - 2x - 3}{x^3 - x} dx = \int 2x + 3 + \frac{A}{x} + \frac{B}{x-1} + \frac{C}{x+1} dx \end{aligned}$$

$$\text{所以：} 9x^2 - 2x - 3 = A(x-1)(x+1) + Bx(x+1) + Cx(x-1)$$

$$\text{對照係數：} \begin{cases} A + B + C = 9, \\ B - C = -2, \\ -A = -3 \end{cases} \Rightarrow (A, B, C) = (3, 2, 4)$$

$$= \int \left(2x + 3 + \frac{3}{x} + \frac{2}{x-1} + \frac{4}{x+1} \right) dx = x^2 + 3x + 3 \ln|x| + 2 \ln|x-1| + 4 \ln|x+1| + C$$

$$\begin{aligned} \blacktriangleright \int \frac{3x^4 + x^3 + 3x^2 - 3x - 2}{x^3 - x} dx \\ = \int 3x + 1 + \frac{6x^2 - 2x - 2}{x^3 - x} dx = \int 3x + 1 + \frac{A}{x} + \frac{B}{x-1} + \frac{C}{x+1} dx \end{aligned}$$

$$\text{所以：} 6x^2 - 2x - 2 = A(x-1)(x+1) + Bx(x+1) + Cx(x-1)$$

$$\text{對照係數：} \begin{cases} A + B + C = 6, \\ B - C = -2, \\ -A = -2 \end{cases} \Rightarrow (A, B, C) = (2, 1, 3)$$

$$= \int \left(3x + 1 + \frac{2}{x} + \frac{1}{x-1} + \frac{3}{x+1} \right) dx = \frac{3}{2}x^2 + x + 2 \ln|x| + \ln|x-1| + 3 \ln|x+1| + C$$

2. 變數變換

將被積函數的一部分打包成一個新變數 $u = u(x)$ ，有機會使積分難度降低。打包時，若成功湊出 $u'(x) dx$ 則積分會變得更加容易。而定積分在使用變數變換時，需要注意上下限範圍。最後記得換回最一開始的變數。

Question Calculate the following integral.

$$\begin{aligned} \triangleright \int 2x\sqrt{x^2+1} dx &= \int \sqrt{u} du, \text{ 令 } u = x^2 + 1, \text{ 則 } du = 2x dx \\ &= \int u^{1/2} du \\ &= \frac{u^{3/2}}{3/2} + C \\ &= \frac{2}{3}(x^2 + 1)^{3/2} + C \end{aligned}$$

$$\begin{aligned} \blacktriangleright \int x^2\sqrt{x^3+1} dx &= \int \sqrt{u} \frac{du}{3}, \text{ 令 } u = x^3 + 1, \text{ 則 } du = 3dx \\ &= \frac{1}{3} \frac{u^{3/2}}{3/2} + C \\ &= \frac{2}{9}(x^3 + 1)^{3/2} + C \end{aligned}$$

$$\begin{aligned} \triangleright \int \sec x dx &= \int \frac{\sec x(\sec x + \tan x)}{\sec x + \tan x} dx \\ &= \int \frac{u'}{u} du, \text{ 令 } u = \sec x + \tan x, \text{ 則 } du = (\sec x \tan x + \sec^2 x) dx \\ &= \ln |u| + C \\ &= \ln |\sec x + \tan x| + C \end{aligned}$$

$$\begin{aligned} \blacktriangleright \int \tan x dx &= \int \frac{\sin x}{\cos x} dx \\ &= \int -\frac{1}{u} du, \text{ 令 } u = \cos x, \text{ 則 } du = -\sin x dx \\ &= -\ln |u| + C \\ &= -\ln |\cos x| + C \end{aligned}$$

Remark 變數變換本質上對應至微分規則的連鎖率。

2.1. 三角代換

三角代換算式變數變換的一種特殊方法，三角代換是為了使用一些三角恆等式將 x 直接替換成 $\sin \theta$ 、 $\tan \theta$ 、 $\sec \theta$ ：

積分中的表達式	代換方法	代換結果
$\sqrt{a^2 - x^2}$	$x = a \sin \theta, -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$	$1 - \sin^2 \theta = \cos^2 \theta$
$\sqrt{a^2 + x^2}$	$x = a \tan \theta, -\frac{\pi}{2} < \theta < \frac{\pi}{2}$	$1 + \tan^2 \theta = \sec^2 \theta$
$\sqrt{x^2 - a^2}$	$x = a \sec \theta, 0 \leq \theta < \frac{\pi}{2} \text{ or } \pi \leq \theta < \frac{3\pi}{2}$	$\sec^2 \theta - 1 = \tan^2 \theta$

Question Calculate the following integral.

$$\begin{aligned}
 \triangleright \int \frac{1}{1+x^2} dx &= \int \cos^2 u \sec^2 u du, \text{ 令 } x = \tan u, \text{ 則 } dx = \sec^2 u du \\
 &= \int 1 du \\
 &= u + C \\
 &= \tan^{-1} x + C
 \end{aligned}$$

$$\begin{aligned}
 \blacktriangleright \int \frac{1}{\sqrt{4-x^2}} dx &= \int \frac{\sec u}{2} \cdot 2 \cos u du, \text{ 令 } x = 2 \sin u, \text{ 則 } dx = 2 \cos u du \\
 &= \int 1 du \\
 &= u + C \\
 &= \sin^{-1} \frac{x}{2} + C
 \end{aligned}$$

Remark 只要記住「根號內的數值必須大於等於零」。若遇到分母有一次式，如 $\int \frac{1}{x^2 + 2x + 5} dx$ ，可先將分母配方後再使用三角代換，如 $\int \frac{1}{(x+1)^2 + 2^2} dx$ 並令 $x+1 = \tan u$ ，則 $dx = \sec^2 u du$ 即完成代換。

3. 分部積分

微分乘法法則告訴我們：

$$\frac{d}{dx}(F(x) \cdot G(x)) = F(x)g(x) + f(x)G(x)$$

其中 $F'(x) = f(x)$ 、 $G'(x) = g(x)$. 同時對兩邊積分可以得到：

$$F(x) \cdot G(x) = \int F(x)g(x) \, dx + \int f(x)G(x) \, dx$$

移項後有：

$$\int F(x)g(x) \, dx = F(x) \cdot G(x) - \int f(x)G(x) \, dx$$

這即是分部積分的基礎，它幫助我們處理有關函數乘積的積分.

Question Calculate the following integrals by parts.

$$\begin{aligned} \triangleright \int x \ln x \, dx &= x^2 \frac{\ln x}{2} - \int \frac{x}{2} \, dx \\ &= \frac{x^2}{2} \ln x - \frac{x^2}{4} + C \end{aligned}$$

$$\begin{aligned} \blacktriangleright \int \ln x \, dx &= x \ln x - \int 1 \, dx \\ &= x \ln x - x + C \end{aligned}$$

$$\begin{aligned} \triangleright \int x e^x \, dx &= x e^x - \int e^x \, dx \\ &= e^x(x - 1) + C \end{aligned}$$

$$\begin{aligned} \blacktriangleright \int x^2 e^x \, dx &= x^2 e^x - \int 2x e^x \, dx \\ &= x^2 e^x - 2(x e^x - \int e^x \, dx) \\ &= e^x(x^2 - 2x + 2) + C \end{aligned}$$

$$\begin{aligned}\triangleright \int e^x \sin x \, dx &= e^x \sin x - \int e^x \cos x \, dx \\ &= e^x \sin x - \left(e^x \cos x - \int e^x (-\sin x) \, dx \right) \\ &= e^x (\sin x - \cos x) - \int e^x \sin x \, dx \\ &\Rightarrow 2 \int e^x \sin x \, dx = e^x (\sin x - \cos x) \\ \therefore \int e^x \sin x \, dx &= \frac{e^x}{2} (\sin x - \cos x) + C\end{aligned}$$

$$\begin{aligned}\blacktriangleright \int e^x \cos x \, dx &= e^x \cos x + \int e^x \sin x \, dx \\ &= e^x \cos x + \left(e^x \sin x - \int e^x \cos x \, dx \right) \\ &= e^x (\sin x + \cos x) - \int e^x \cos x \, dx \\ &\Rightarrow 2 \int e^x \cos x \, dx = e^x (\sin x + \cos x) \\ \therefore \int e^x \cos x \, dx &= \frac{e^x}{2} (\sin x + \cos x) + C\end{aligned}$$

Remark 可以發現，分部積分使用時機大致有：

- 多項式乘上指數函數、對數函數、三角函數、反三角函數，並且微分對象優先嘗試順序為對數、反三角、多項式、三角、指數。
- 某函數微分有循環的，如 e^x 、 $\sin(x)$ 、 $\cos(x)$ ，有機會可以回到原積分式。

4. 瑕積分

Definition 若積分上、下限中有一端為無窮，則稱該積分屬於第一類瑕積分：

- $\int_a^\infty f(x) \, dx = \lim_{t \rightarrow \infty} \int_a^t f(x) \, dx$
- $\int_{-\infty}^b f(x) \, dx = \lim_{t \rightarrow -\infty} \int_t^b f(x) \, dx$
- $\int_{-\infty}^\infty f(x) \, dx = \int_{-\infty}^a f(x) \, dx + \int_a^\infty f(x) \, dx$

Remark 對應的極限若存在，稱該積分收斂 (Convergent)；若不存在，則稱發散 (Divergent)。

Question Calculate the following improper integrals.

$$\begin{aligned} \triangleright \int_1^\infty 2x^{-3} \, dx &= \lim_{t \rightarrow \infty} \int_1^t 2x^{-3} \, dx \\ &= \lim_{t \rightarrow \infty} [-x^{-2}]_1^t \\ &= \lim_{t \rightarrow \infty} (-t^{-2} + 1) = 1 \end{aligned}$$

$$\begin{aligned} \blacktriangleright \int_{-\infty}^{-1} \frac{1}{\sqrt[3]{x}} \, dx &= \lim_{t \rightarrow -\infty} \int_t^{-1} x^{-1/3} \, dx \\ &= \lim_{t \rightarrow -\infty} \left[\frac{3}{2} x^{2/3} \right]_t^{-1} \\ &= \frac{3}{2}(-1)^{2/3} - \lim_{t \rightarrow -\infty} \frac{3}{2} t^{2/3} \rightarrow -\infty \quad \Rightarrow \text{該積分發散 (Diverges)} \end{aligned}$$

$$\begin{aligned} \triangleright \int_{-\infty}^\infty \frac{1}{x^2 + 1} \, dx &= \int_{-\infty}^0 \frac{1}{x^2 + 1} \, dx + \int_0^\infty \frac{1}{x^2 + 1} \, dx \\ &= \lim_{t \rightarrow -\infty} \int_t^0 \frac{1}{x^2 + 1} \, dx + \lim_{s \rightarrow \infty} \int_0^s \frac{1}{x^2 + 1} \, dx \\ &= \lim_{t \rightarrow -\infty} (\tan^{-1} 0 - \tan^{-1} t) + \lim_{s \rightarrow \infty} (\tan^{-1} s - \tan^{-1} 0) \\ &= \frac{\pi}{2} + \frac{\pi}{2} = \pi \end{aligned}$$

$$\begin{aligned} \blacktriangleright \int_{-\infty}^\infty \frac{1}{x^2 + 9} \, dx &= \int_{-\infty}^0 \frac{1}{x^2 + 9} \, dx + \int_0^\infty \frac{1}{x^2 + 9} \, dx \\ &= \lim_{t \rightarrow -\infty} \int_t^0 \frac{1}{x^2 + 9} \, dx + \lim_{s \rightarrow \infty} \int_0^s \frac{1}{x^2 + 9} \, dx \\ &= \lim_{t \rightarrow -\infty} \frac{1}{3} (\tan^{-1}(0/3) - \tan^{-1}(t/3)) + \lim_{s \rightarrow \infty} \frac{1}{3} (\tan^{-1}(s/3) - \tan^{-1}(0/3)) \\ &= \frac{1}{3} \left(\frac{\pi}{2} + \frac{\pi}{2} \right) = \frac{\pi}{3} \end{aligned}$$

Definition 若被積函數在積分區間內某點不連續或無界，則稱該積分屬於**第二類瑕積分**：

- $\int_a^b f(x) \, dx = \lim_{t \rightarrow b^-} \int_a^t f(x) \, dx$ (當 $f(x)$ 在 $x = b$ 不連續)
- $\int_a^b f(x) \, dx = \lim_{t \rightarrow a^+} \int_t^b f(x) \, dx$ (當 $f(x)$ 在 $x = a$ 不連續)
- $\int_a^b f(x) \, dx = \int_a^c f(x) \, dx + \int_c^b f(x) \, dx$ (當 $f(x)$ 在 (a, b) 內某點 c 不連續)

Question Calculate the following improper integrals.

$$\begin{aligned}
 \triangleright \int_0^1 \frac{1}{x} \, dx &= \lim_{t \rightarrow 0^+} \int_t^1 \frac{1}{x} \, dx \\
 &= \lim_{t \rightarrow 0^+} [\ln x]_{x=t}^{x=1} \\
 &= \lim_{t \rightarrow 0^+} \ln 1 - \ln t \\
 &= \lim_{t \rightarrow 0^+} -\ln t \rightarrow \infty \quad \Rightarrow \text{該積分發散 (Diverges)}
 \end{aligned}$$

$$\begin{aligned}
 \blacktriangleright \int_{-1}^0 \frac{1}{x^2} \, dx &= \lim_{t \rightarrow 0^-} \int_{-1}^t x^{-2} \, dx \\
 &= \lim_{t \rightarrow 0^-} [-x^{-1}]_{x=-1}^{x=t} \\
 &= \lim_{t \rightarrow 0^-} (-t^{-1}) - ((-1)^{-1}) \\
 &= \lim_{t \rightarrow 0^-} -t^{-1} + 1 \rightarrow \infty \quad \Rightarrow \text{該積分發散 (Diverges)}
 \end{aligned}$$

Theorem. (直接比較審斂法 (Direct Comparison Theorem)) 給定連續函數 $f(x)$ 、 $g(x)$ ，對於任意 $x \geq a$ 有 $f(x) \geq g(x) \geq 0$ ，那麼：

1. 若 $\int_a^\infty f(x) \, dx$ 收斂，則 $\int_a^\infty g(x) \, dx$ 收斂
2. 若 $\int_a^\infty g(x) \, dx$ 發散，則 $\int_a^\infty f(x) \, dx$ 發散

Question Determine whether the following integrals converge.

$$\begin{aligned}
 \triangleright \int_1^\infty \frac{\sin^2 x}{x^2} \, dx & \text{ (Hint: } 0 \leq \sin^2 x \leq 1, \forall x \in \mathbb{R} \text{)} \\
 & \text{由於 } 0 \leq \sin^2 x \leq 1, \text{ 我們有 } 0 \leq \frac{\sin^2 x}{x^2} \leq \frac{1}{x^2}, \quad \forall x \geq 1 \\
 & \text{並且積分 } \int_1^\infty \frac{1}{x^2} \, dx = 1 \text{ 收斂} \\
 & \text{由直接比較審斂法, } \int_1^\infty \frac{\sin^2 x}{x^2} \, dx \text{ 亦收斂.}
 \end{aligned}$$

習題

► 積分技巧與基本計算

1. (師大微乙 (一)114#3) Evaluate the integrals. (計算積分)

(a) $\int_0^{\pi/2} \cos^3(3\theta) \sin^2(6\theta) d\theta$

(e) $\int \sqrt[3]{\tan x} dx$

(b) $\int x^3 \sin(2x) dx$

(f) $\int \ln(x) \cdot x^{\ln(x)-1} dx$

(c) $\int \frac{2x^2 - 6x + 4}{(x^2 + 1)(x - 1)^3} dx$

(g) $\int_0^{2025} \frac{\sqrt{x}}{\sqrt{x} + \sqrt{2025 - x}} dx$

(d) $\int \tan^{-1}(\sqrt{x}) dx$

(h) $\int \frac{\sin^2(x)}{1 + \cos^2(x)} dx$

2. (師大微乙 (一)113 暑 #2) Evaluate the following integrals.

(a) $\int \left(x + \sqrt{x} + 1 + \frac{1}{\sqrt{x}} + \frac{1}{x} \right) dx$

(d) $\int \frac{2x}{\sqrt{1+x^4}} dx$

(b) $\int (x^2 \sin x + \cos^2 x) dx$

(e) $\int \frac{e^x}{e^{2x} - 3e^x + 2} dx$

(c) $\int (\tan^2 x + \tan^4 x) dx$

(f) $\int_{-1}^1 (x^3 + 3)\sqrt{4 - x^2} dx$

3. (師大微乙 (一)113 暑 #3) Let a be a real number and $a > 0$.

(a) Evaluate $\int x \ln x dx$

(b) If a satisfies the equation $\int_0^a x \ln x dx = 0$, find the value of a .

4. (師大微乙 (一)113#2) Evaluate the integrals.

(a) $\int \frac{\cos^5 x}{\sin^2 x} dx$

(d) $\int \frac{\tan x (\sec^2 x + 4 \sec x)}{\sec^2 x + 5 \sec x - 6} dx$

(b) $\int \frac{x^2}{\sqrt{49 - x^2}} dx$

(e) $\int_{-1}^1 \sqrt{x^{10} + x^2} \sec x \tan x + \frac{1}{1 + x^2} dx$

(c) $\int \frac{2x^3 + 4}{(x - 1)^2(x^2 + 1)} dx$

5. (師大微乙 (一)113#6)

(a) Evaluate $\int_0^1 \ln x dx$. (Note: this integral is an improper integral(瑕積分).)

(b) Let a be a real number and $a > 0$. If a satisfies the equation $\int_0^a \ln x dx = 0$, find a .

6. (師大微乙 (一)112#2) Evaluate the following integrals.

(a) $\int \frac{x^2}{x^4 - 1} dx$

(d) $\int \sec^5 x dx$

(b) $\int_0^{\pi^2/4} \cos \sqrt{x} dx$

(e) $\int \frac{(1-y^2)^{5/2}}{y^8} dy$

(c) $\int_0^e \ln x dx$

7. (師大微乙 (一)111#1) Evaluate the integrals.

(a) $\int \frac{\ln(\ln x)}{x \ln x} dx$

(e) $\int \sqrt{\sin x} \cos^3 x dx$

(b) $\int_0^1 x \ln x dx$

(f) $\int \frac{x e^{2x}}{(2x+1)^2} dx$

(c) $\int \frac{x^3}{x^2 + 2x + 1} dx$

(g) $\int \frac{\sec^2 \theta}{\tan^3 \theta - \tan^2 \theta} d\theta$

(d) $\int \frac{x}{\sqrt{1+x^4}} dx$

(h) $\int_{-\infty}^1 \frac{9r^2}{\sqrt{1-r^3}} dr$

8. (師大微乙 (一)110#1) 計算下列積分. (Evaluate the following integrals.)

(a) $\int_1^4 \left(\frac{2}{x^3} - \sqrt{x} \right) dx$

(e) $\int e^x \cos 2x dx$

(b) $\int x^2 \sqrt{x^3 + 1} dx$

(f) $\int \frac{2}{16x^2 - 1} dx$

(c) $\int_0^3 \frac{x}{\sqrt{x+1}} dx$

(g) $\int \sqrt{9-x^2} dx$

(d) $\int \cos \sqrt{x} dx$

(h) $\int_4^6 \frac{1}{\sqrt{8x-x^2}} dx$

9. (師大微乙 (一)110#5) 計算以下瑕積分. (Evaluate the following improper integrals.)

(a) $\int_0^{\infty} x e^{-2x} dx$

(b) $\int_0^1 \ln x dx$

10. (師大微乙 (一)109#1) Evaluate the integrals.

(a) $\int \sqrt{16-4x^2} dx$

(e) $\int \frac{\sqrt{x^2+3}}{x} dx$

(b) $\int_1^{\infty} \frac{1}{x(\ln x)^3} dx$

(f) $\int \sec(2x) \tan^2(2x) dx$

(c) $\int x^4 \cos(2x) dx$

(g) $\int_0^{\pi/2} \frac{\cos(\theta)}{(\sin^2(\theta) + 4 \cos^2(\theta))^{3/2}} d\theta$

(d) $\int \sin(7x) \cos(3x) dx$

(h) $\int \frac{5x^2 + 20x + 6}{x^3 + 2x^2 + x} dx$

11. (師大微乙 (一)108#3)

(a) Does the improper integral $\int_1^{\infty} \frac{\ln t}{2\sqrt{t}} dt$ converge or diverge? Give reasons for your answer.

(b) Use the part (a) to evaluate $\lim_{x \rightarrow \infty} \frac{\int_1^x \frac{\ln t}{2\sqrt{t}} dt}{\sqrt{x^3}}$.

12. (師大微乙 (一)108#4) Find each integral.

(a) $\int \frac{8x}{x^3 + x^2 - x - 1} dx$

(d) $\int e^x \cos x dx$

(b) $\int \tan^4(5x) dx$

(e) $\int_0^{\pi/2} \frac{\cos^5 x}{\sin^5 x + \cos^5 x} dx$ (Hint: Use the substitution $u = \frac{\pi}{2} - x$.)

(c) $\int e^x \sqrt{1 - e^{2x}} dx$

13. (師大微乙 (一)107#1) Evaluate the integrals.

(a) $\int \frac{2x+1}{\sqrt{4-x}} dx$

(e) $\int \frac{x}{x^2 - 2x + 3} dx$

(b) $\int \frac{1}{\sqrt{x}(1+\sqrt{x})} dx$

(f) $\int \tan x \cdot \cos(2x) dx$

(c) $\int x e^{2x} dx$

(g) $\int_0^{\sqrt{3}} \frac{x^2}{(4-x^2)^{3/2}} dx$

(d) $\int \frac{-x^3 + 5x^2 - 4x + 9}{x^4 - 2x^3 + 3x^2 - 4x + 2} dx$

(h) $\int \sqrt{1+x^{-1}} dx$

14. (師大微乙 (一)107#5) Find the improper integral

$$\int_0^4 \frac{\ln x}{\sqrt{x}} dx.$$

15. (師大微乙 (一)106#1) Evaluate the integrals.

(a) $\int (x+1)5^{(x+1)^2} dx$

(e) $\int \sec^4 x \tan^3 x dx$

(b) $\int \frac{2x^3 - 4x^2 - 15x + 5}{x^2 - 2x - 8} dx$

(f) $\int \frac{\sin x}{\cos x - \cos^2 x} dx$

(c) $\int e^{\sqrt{2x}} dx$

(g) $\int \frac{\ln \sqrt{x}}{x} dx$

(d) $\int e^x \sqrt{1 - e^{2x}} dx$

(h) $\int_0^1 x \ln x dx$

16. (師大微乙 (一)106#5) 給定下列之瑕積分 (improper integral), 請做以下兩小題.

$$\int_1^{\infty} \left(\frac{c}{x^2} - \frac{1}{3x} \right) dx$$

(a) 找出使得上述瑕積分收斂 (converge) 的 c 值.

(b) 求 (a) 答案 c 值對應之瑕積分值.

17. (師大微乙 (一)105#1) Evaluate the integrals.

(a) $\int_1^4 (3 - |x - 3|) \, dx$

(e) $\int \frac{\ln x}{x^3} \, dx$

(b) $\int \frac{3}{x^2 + 3x - 10} \, dx$

(f) $\int \frac{\sqrt{x^2 - 25}}{x} \, dx$

(c) $\int \sin^2 \theta \tan \theta \sqrt{\cos \theta} \, d\theta$

(g) $\int \frac{x^3 e^{x^2}}{(x^2 + 1)^2} \, dx$

(d) $\int \frac{\sec x \tan x}{2 \sec x - 1} \, dx$

(h) $\int_0^2 \frac{1}{\sqrt[3]{x-1}} \, dx$

18. (師大微乙 (一)104#1) 計算下列積分：

(a) $\int_0^1 (x^3 + 2x) \, dx$

(e) $\int \sec^3(z) \, dz$

(b) $\int \frac{2x + 1}{x^2 - 7x + 12} \, dx$

(f) $\int_1^9 \frac{\log_3 x}{x} \, dx$

(c) $\int_0^{\pi/2} \cos^3(x) \, dx$

(g) $\int \sin \sqrt{2x} \, dx$

(d) $\int_1^e \frac{dy}{y(1 + (\ln y)^2)}$

(h) $\int \frac{ds}{1 + e^s}$

19. (師大微乙 (一)104#5) 判斷下列瑕積分收斂或發散 (請標明使用何種檢定法).

$$\int_1^\infty \frac{1}{\sqrt{x^6 + 1}} \, dx$$

20. (師大微乙 (一)103#1) 底下共有 8 題積分計算題

(a) $\int_1^4 x^2 + 3x + 1 \, dx$

(e) $\int \frac{\sqrt{1-x^2}}{x} \, dx$

(b) $\int \frac{1}{x^2 - 5x + 6} \, dx$

(f) $\int \ln x \, dx$

(c) $\int \sin^3 x \cos^4 x \, dx$

(g) $\int \frac{\ln x}{x} \, dx$

(d) $\int \tan^2 x \sec^3 x \, dx$

(h) $\int \frac{2z}{\sqrt[3]{z^2 + 1}} \, dz$

21. (師大微乙 (一)103#5)

(a) 驗算 $\int_0^2 x^2 \, dx + \int_0^4 \sqrt{x} \, dx = 8$

(b) 求 $\int_0^{\frac{\pi}{2}} \sin x \, dx + \int_0^1 \sin^{-1} x \, dx$

(d) 判斷下列瑕積分收斂或發散，若收斂請計算其值。

$$\int_0^{\infty} \frac{e^{\tan^{-1} x}}{1+x^2} \, dx$$

22. (師大微乙 (一)102#1) 計算下列積分:

(a) $\int_0^1 (x^2 + 3x + 2) \, dx$

(e) $\int \tan^3 x \sec^4 x \, dx$

(b) $\int \frac{1}{x^2 + 3x + 2} \, dx$

(f) $\int \frac{1 + \sin x}{\cos^2 x} \, dx$

(c) $\int_0^1 x\sqrt{1-x} \, dx$

(g) $\int x^2 e^{-x} \, dx$

(d) $\int_0^{2/3} \sqrt{4-9x^2} \, dx$

(h) $\int \sin \sqrt{x} \, dx$

23. (師大微乙 (一)102#5) 判斷下列瑕積分收斂或發散，若收斂，計算其值。

(a) $\int_0^{\infty} \frac{1}{x^2 + 1} \, dx$

(b) $\int_0^{\infty} \frac{1}{\sqrt{x} + 1} \, dx.$